sEMG Gesture Classifier ASIC

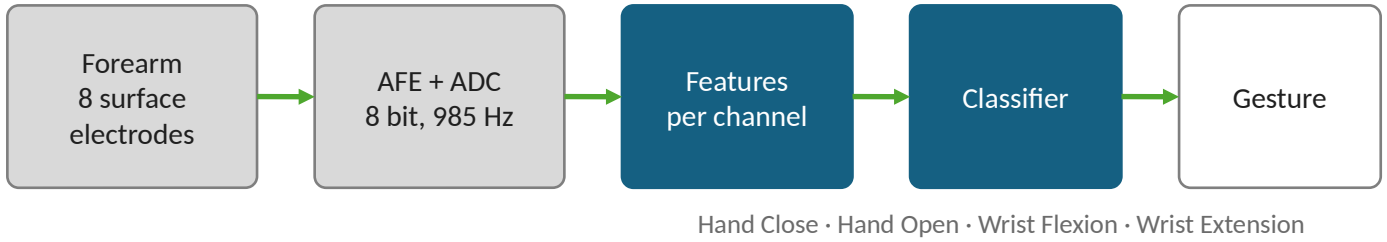
Subject-independent hand-gesture recognition on chained INT8 LDA boards

Vinay Hariharan, Sudhindra Sahoo, Yashvardhan Khandegar, Digvijay Kumbhar

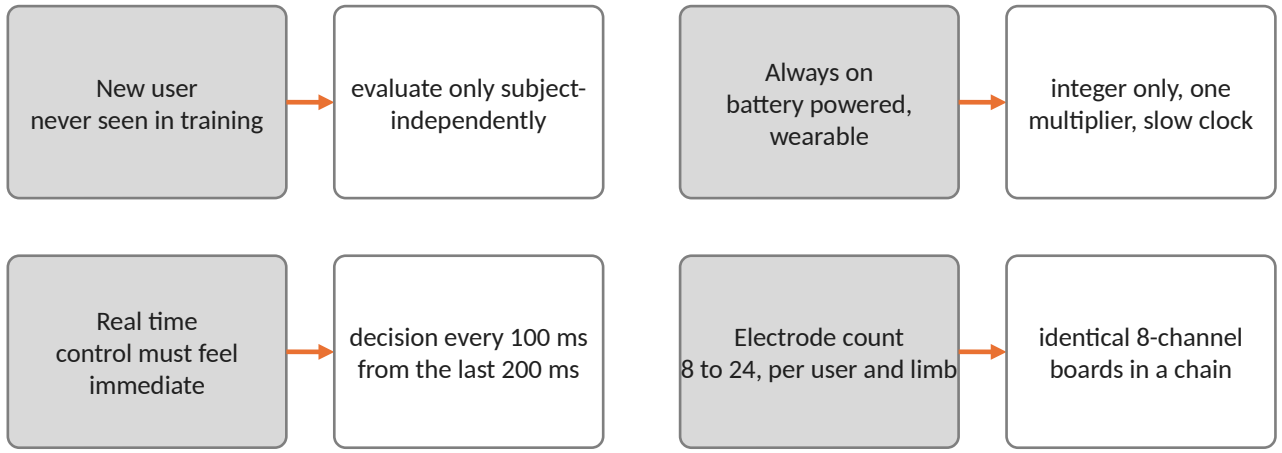
Under the supervision of Prof. Laxmeesha Somappa · BioNICS, Department of Electrical Engineering, IIT Bombay

PROBLEM

Recognising Hand Gestures from Forearm Muscle Signals



What makes it hard → what it forces on the design



SIGNAL

Surface EMG: forearm muscle activity measured on the skin. Each gesture uses a different set of muscles.

GOAL

Decide the gesture continuously on a wearable, e.g. a prosthetic hand or a gesture interface.

HARD PART

A new user's signal differs: skin, fat, electrode placement and physiology all change it.

TARGET

≥ 60 % accuracy on users never seen in training, with the least hardware.

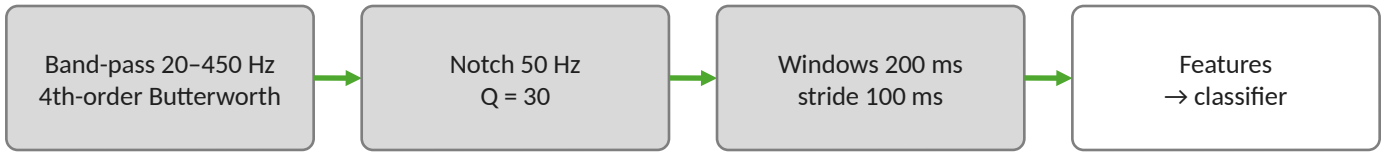
Every design choice is judged by accuracy on an unseen person and by hardware cost.

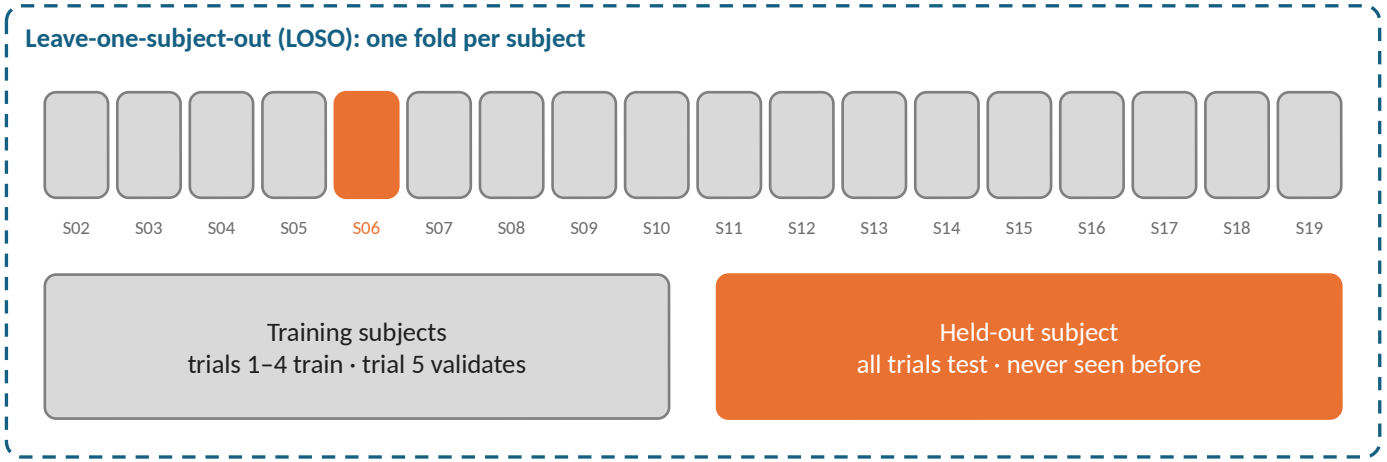
DATASET

FORS-EMG and How We Evaluate

FORS-EMG [1]: 19 subjects × 12 gestures × 3 forearm orientations × 5 trials
8 electrodes · 985 Hz · 8000 samples (8.1 s) per recording

causal filters, as the hardware runs them





DATA

Public FORS-EMG [1]: each gesture recorded at rest, in supination and in pronation; a class pools all three.

PRE-PROCESSING

Causal band-pass and notch filters, exactly as a chip would run them.

EVALUATION

Leave-one-subject-out: the test subject never touches training, normalisation or quantisation.

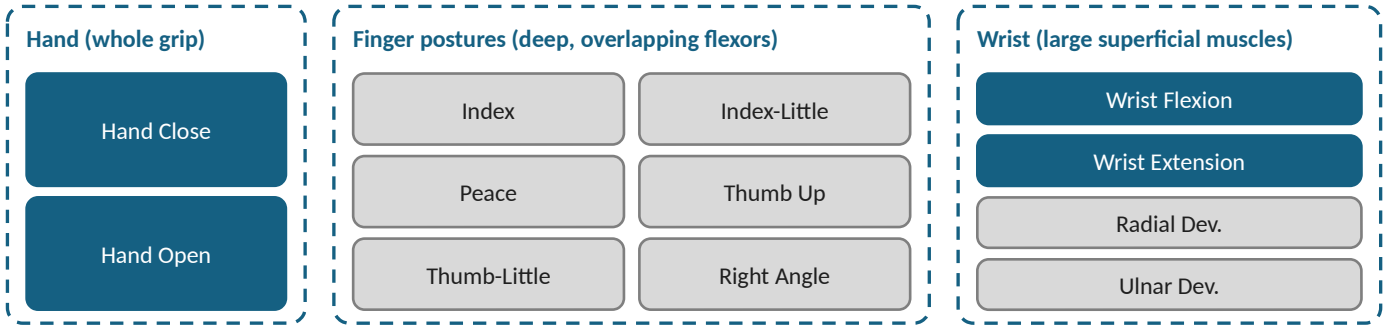
STATISTICS

Paired per-subject tests; a difference counts only if its 95 % CI excludes zero.

Only subject-independent accuracy is reported: the number a new user would actually see.

DATASET

Mixing Gesture Kinds: Which, and How Many



dark blue: the chosen set B (2 hand + 2 wrist)

Set	Mix of kinds	Gestures	INT8 %	Chance
A (4)	hand + finger	Hand Close, Hand Open, Index, Index-Little	65.3	25.0
B (4)	**hand + wrist**	**Hand Close, Hand Open, Wrist Flexion, Wrist Extension**	**79.4**	**25.0**
D (6)	hand + finger + wrist	Hand Close, Hand Open, Peace, Right Angle, Radial Deviation, Wrist Flexion	69.0	16.7
C (12)	everything	all 12 gestures	48.8	8.3

THREE KINDS

Hand and wrist movements use large, superficial muscles; finger postures use deep, overlapping flexors.

HAND + FINGER (A)

Stuck at ≈ 65 % for every model; Index ↔ Index-Little alone caused ≈ 40 % of the errors.

HAND + WRIST (B)

79.4 %: +14.1 pp ($p = 3.8 \times 10^{-9}$), the largest gain in the project.

HOW MANY

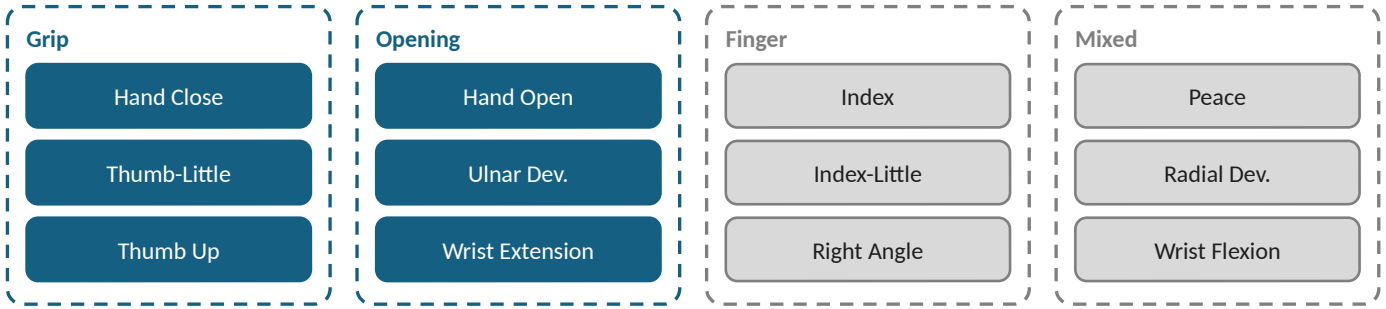
Best 6 of 924 subsets: 69.0 %. All 12 gestures: 48.8 %.

Which gestures we ask for matters more than any model we tried.

DATASET

Merge All 12 Gestures into 4 Classes, or Drop 8?

Most frequent grouping, chosen from the training subjects inside each fold



Hand Open + Wrist Extension together in 19 / 19 folds; Hand Close + both thumb gestures in 17 / 19

19 subjects, chip INT8 LDA	Gestures	Class rows	Accuracy %	vs drop
Drop to HC, HO, WF, WE	**4 of 12**	**4**	**77.82 ± 4.10**	**—**
Merge into 4 groups of 3	all 12	4	60.91 ± 4.55	−16.9 pp
12-class LDA, mapped to groups	all 12	12	63.07 ± 5.07	−14.8 pp
12-class LDA (reference)	all 12	12	46.41 ± 5.55	—
Earlier 6-group map, 6-class LDA	all 12	6	72.38 ± 4.11	−5.5 pp
Earlier 6-group map, 12-class mapped	all 12	12	70.71 ± 4.62	−7.1 pp

QUESTION

Keep all 12 gestures by merging them into 4 classes of 3?

FAIR GROUPING

Best of the 15,400 possible splits, chosen from the training subjects only, inside every fold.

RESULT

60.9 % against 77.8 %: worse on all 19 subjects. Classifying 12 and mapping to groups: 63.1 %.

EARLIER 6-GROUP MAP

72.4 %, still 5.5 pp below dropping (18 of 19 subjects), and one group holds half of all windows.

WHY

Finger gestures stay confusable in any grouping; a merged class mixes patterns one Gaussian (LDA) fits poorly.

We drop gestures: from here on, 4 classes (Hand Close, Hand Open, Wrist Flexion, Wrist Extension).

MODEL SELECTION

Eight Classifiers on the Same 64 Features

Model	FP32 %	INT8 %	MACs	Bytes	vs MLP
MLP 64-16-4	79.65	79.57	1088	1168	—
Gradient boosting	79.63	79.26	0	19,200	tied, over budget
Logistic regression	79.33	79.19	256	272	tied (p = 0.44)
Linear SVM	79.31	78.30	256	272	tied (p = 0.29)
LDA	**77.91**	**77.87**	**256**	**272**	**tied (p = 0.09)**
Random forest	77.84	77.43	0	53,145	worse (p = 0.046)
Extra trees	76.29	76.04	0	38,015	worse (p = 0.0009)
Decision tree	73.69	73.17	0	5,055	worse (p = 0.0002)

MLP width (set A)	64-4	64-16-4	64-32-4	64-64-4	64-128-4
INT8 %	65.28	63.63	63.93	64.87	64.02
Bytes	272	1168	2320	4624	9232

SAME INPUTS

64 features (8 time-domain types × 8 channels), same folds, same INT8 scheme, 10 KB budget.

RESULT

MLP, logistic regression, linear SVM and LDA are statistically tied.

TREES

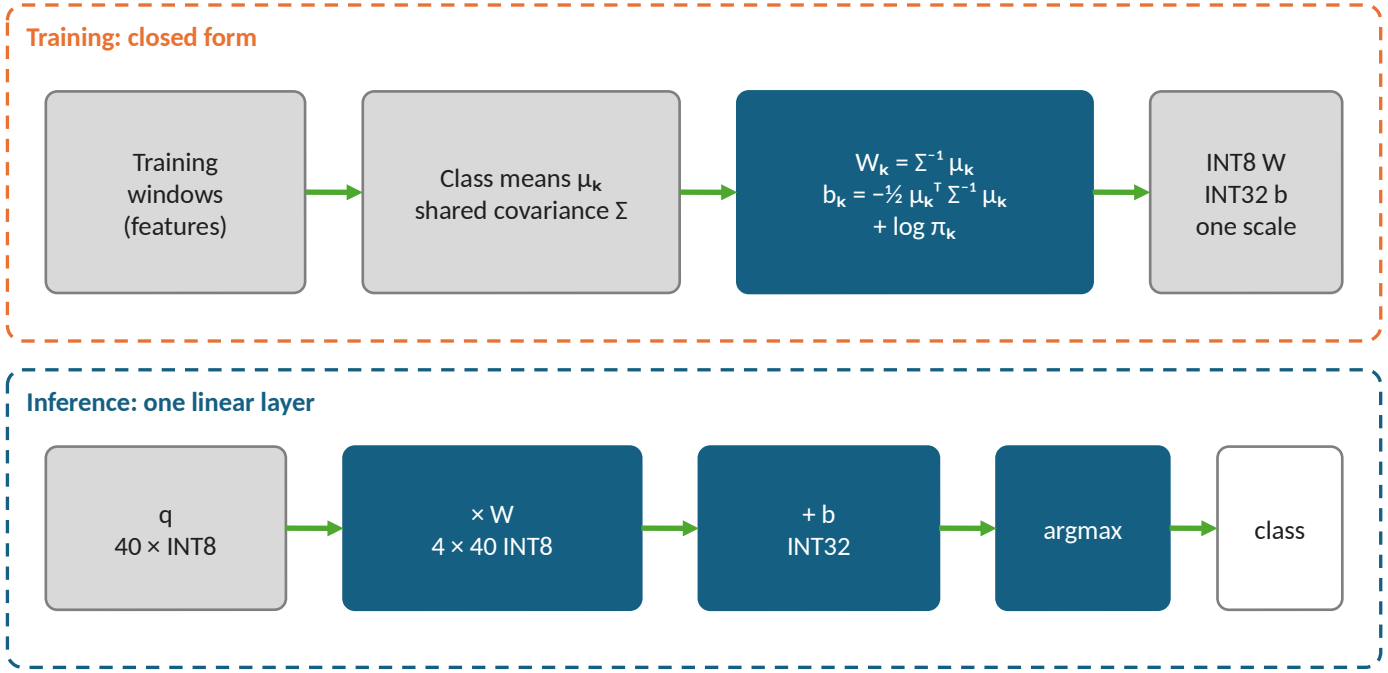
Worse, or far over the memory budget.

BIGGER MLP

34× the memory buys nothing (bottom table): the spread is smaller than any one model's CI.

Capacity is not the bottleneck: the limit is the information in 8 surface channels.

Why LDA, and Why INT8 Costs Nothing



64 features, 4 classes	LDA	MLP 64-16-4
MACs per decision	256	1088
Bytes	272	1168
INT8 accuracy	77.87 %	79.57 %
Training	closed form	gradient descent

CLOSED-FORM TRAINING

Class means and one shared covariance, no optimiser: adapting to a new user is a few matrix operations.

CHEAP

4.3× fewer MACs and bytes than the MLP, for 1.7 pp inside the statistical tie.

INT8 IS FREE

28 FP32-vs-INT8 comparisons: 0.18 pp mean loss, worst 1.01 pp.

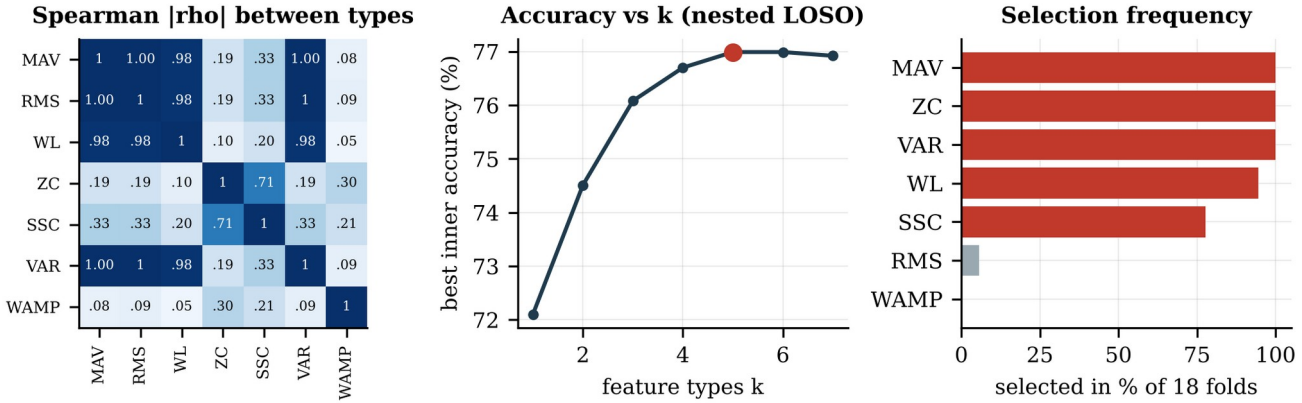
NO RESCALING

One weight scale for all classes leaves the argmax unchanged, so the chip never requantises.

LDA: in the same accuracy tie as an MLP, at a quarter of the cost, and trainable in closed form.

FEATURES

The Features Carry the Information



Type	Per window	Decision
MAV	**mean |x|**	**kept**
WL	**Σ $\|x_n - x_{n-1}\|$**	**kept**
ZC	**number of sign changes**	**kept**
SSC	**number of slope sign changes**	**kept**
VAR	**mean x^2**	**kept**
RMS	√(mean x^2)	dropped: same as MAV / VAR
WAMP	count of $\|\Delta x\|$ > threshold	dropped: saturated (98 %)
IEMG	Σ |x| = N · MAV	dropped: identical LDA

WHY TIME-DOMAIN

sEMG is noise-like: the information is in each channel's amplitude and rate of change. TD features + LDA is the standard real-time scheme [2, 3].

REDUNDANCY

MAV, RMS, WL and VAR are almost the same signal ($|\rho| \geq 0.977$), as also reported in [4].

CULLING

Nested LOSO over all 127 subsets of 7 types; keep the smallest within one standard error of the best.

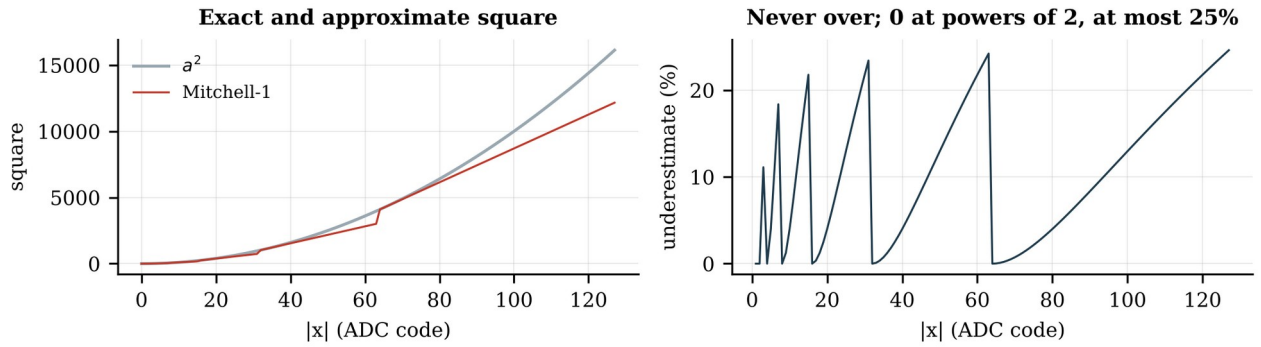
RESULT

5 types: 77.08 % vs 77.31 % with all 8 ($p = 0.20$), with 40 instead of 64 inputs.

Five feature types keep all the accuracy with 40 instead of 64 inputs.

Integer Front End: Add, Compare and Shift Only

Float	On the chip	Hardware
MAV	SABS = Σ |x| (1/N absorbed by the normaliser)	add
WL	$\Sigma \lvert x_n - x_{n-1} \rvert$	subtract, add
ZC	sign of x_n XOR sign of x_{n-1}	XOR, count
SSC	strict sign change of consecutive differences	compare; 2-bit history
VAR	POW = Σa^2, $a^2 \approx a \cdot 2^{(p+1)} - 2^{(2p)}$, $p = \lfloor \log_2 a \rfloor$ [6]	leading-one, 2 shifts, subtract
z-score	q = clip(((x − μ) + 2^(k−1)) >> k, −128, 127)	subtract, add, shift



GOAL

No division, square root, log or float threshold on the chip.

POWER FEATURE

Mitchell square [6]: never overestimates; the window sum has r = 0.9999 with Σx^2.

NORMALISER

(x − μ + 2^(k−1)) >> k, with integer μ and k from the training data.

RESULT

77.39 % vs 77.45 % for float features. Without a power feature: −1.9 pp.

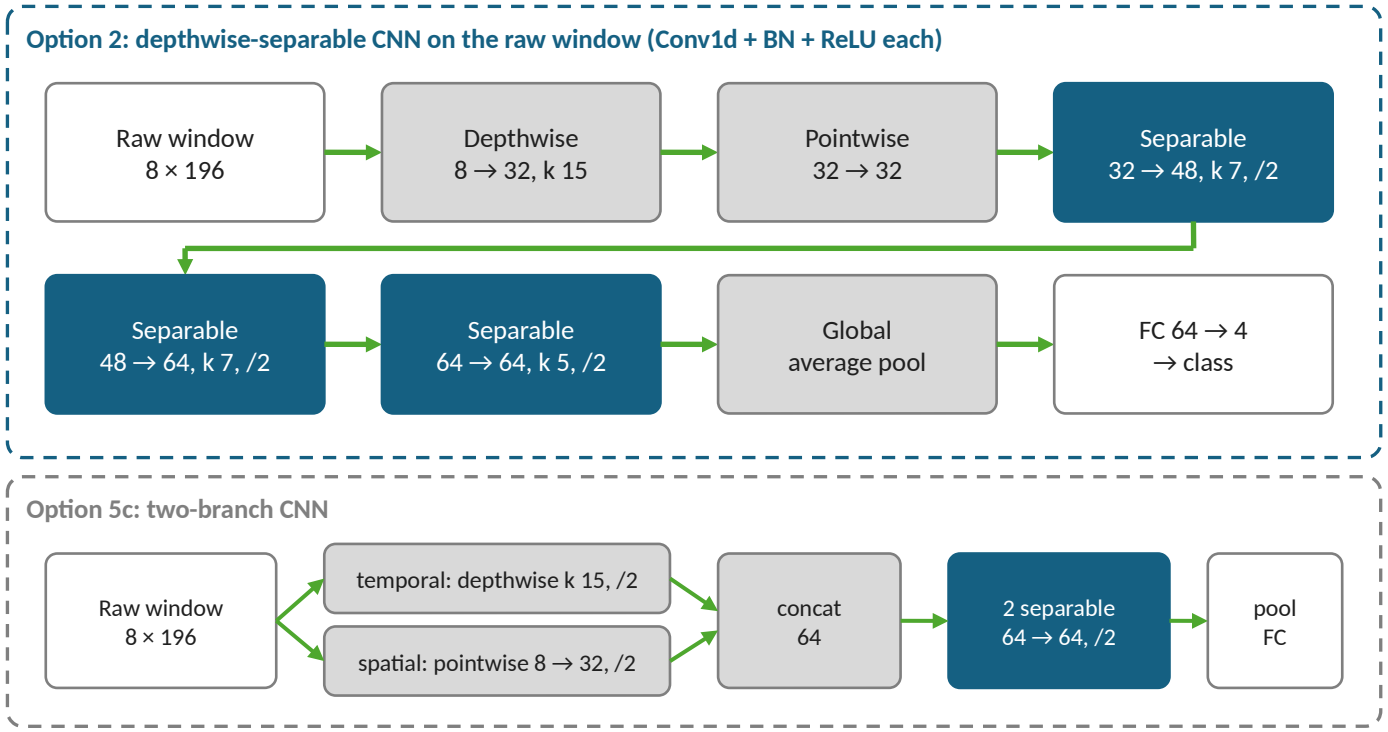
CHIP MODEL

Bit-exact, 8-bit ADC: 77.61 % vs 77.83 % in software (p = 0.49).

The whole front end is add, compare and shift, at no measurable cost in accuracy.

MODEL SELECTION

If Features Matter, Should a CNN Do Better?



19 subjects, INT8	MACs / decision	Parameters	Accuracy
LDA, 40 features	**160**	**164**	**77.79 ± 4.39 %**
CNN, option 2	751,584	11,732	79.10 ± 4.10 %
CNN, option 5c	594,400	12,132	80.48 ± 3.81 %

IDEA

Let a CNN learn its own features from the raw window.

TRAINING

Same windows and folds as the LDA; per-electrode gain augmentation; INT8 with batch-norm folded in.

SEPARABLE CNN

79.1 %: +1.3 pp over the LDA, not significant ($p = 0.20$).

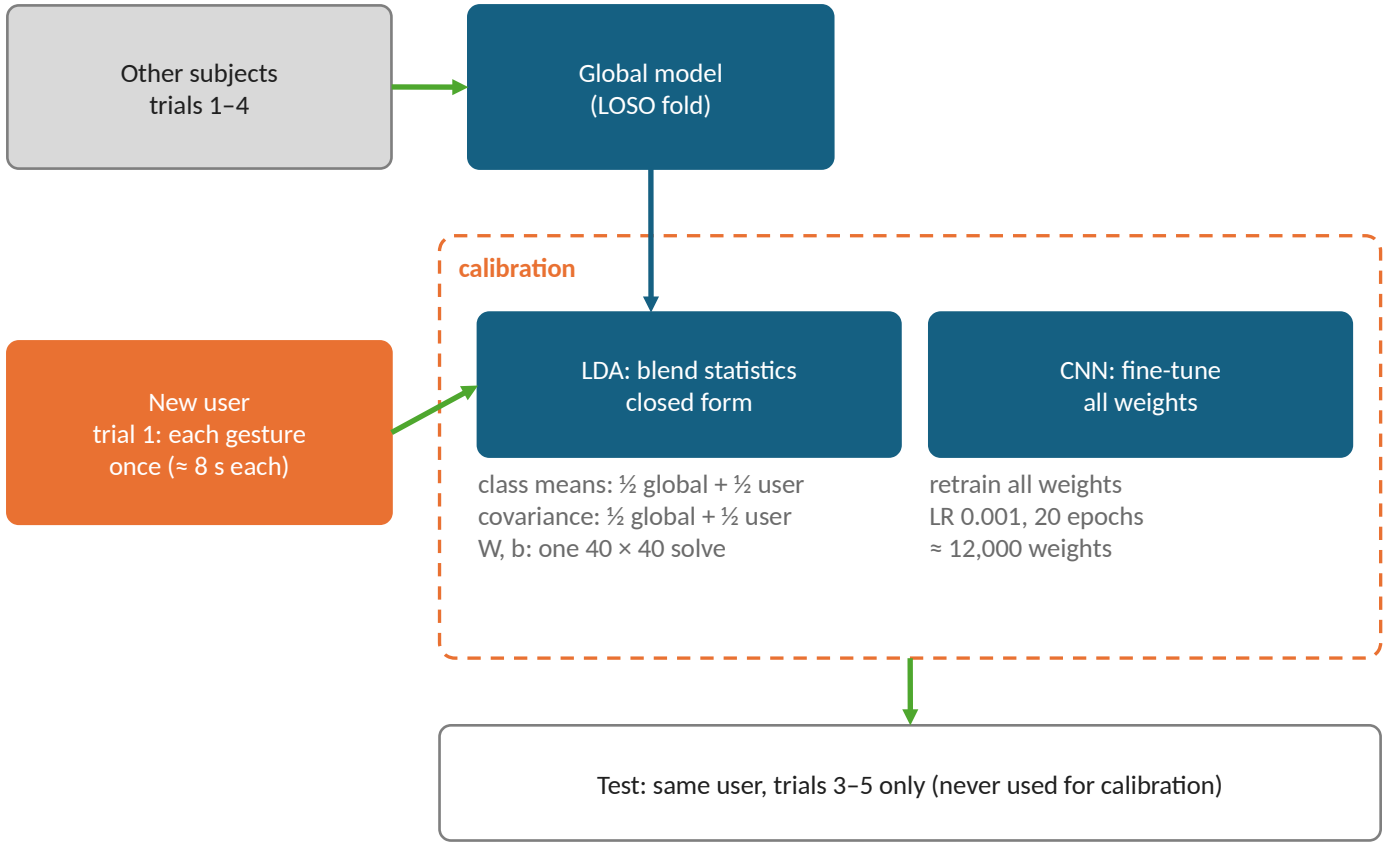
TWO-BRANCH CNN

80.5 %: +2.7 pp ($p = 0.008$), better on 15 of 19 subjects.

The best CNN gains 2.7 pp for ≈ 3,700× the MACs: learned features find little the TD features miss.

CALIBRATION

Adapting to a New User with One Recording per Gesture



PROBLEM

The global model: ≈ 78 % on average, under 60 % for the worst user.

CALIBRATION

The user records each gesture once (≈ 8 s each) and the model adapts.

LDA

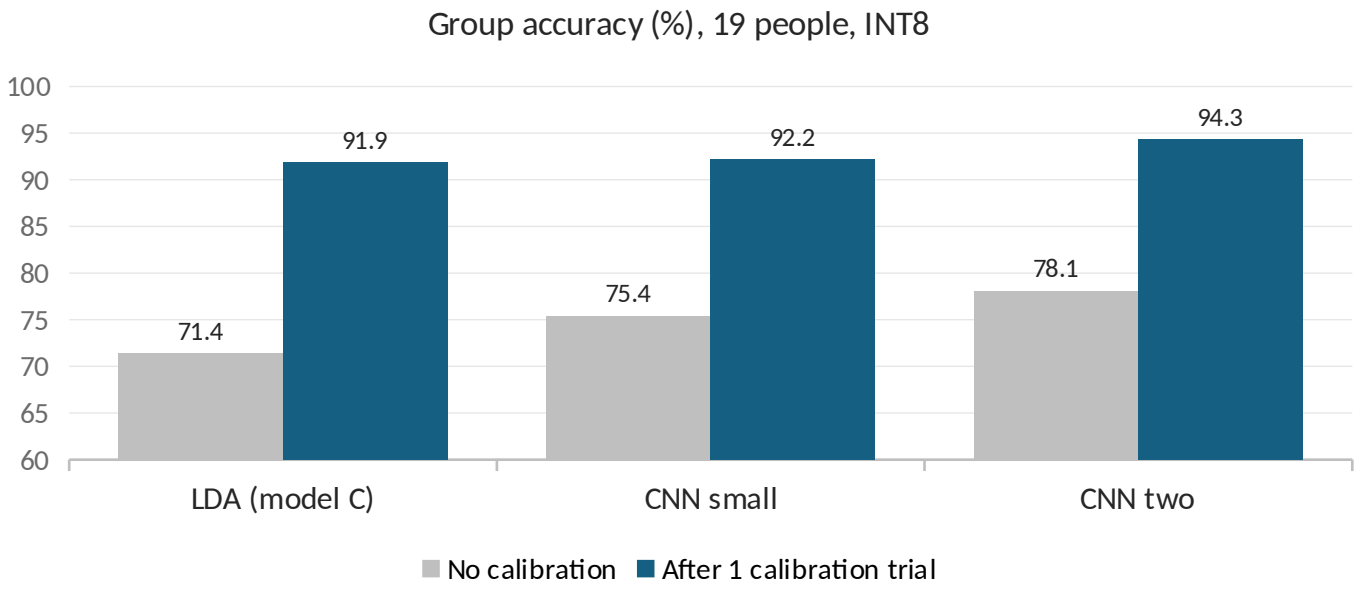
Blend the user's class means and covariance into the global ones, recompute W, b and reload them over the config port.

LDA RESULT

78.0 % → 86.4 % (+8.3 pp); all 18 users improve, the worst most (S15: +30 pp).

One recording per gesture is worth more than any model change we tried.

Calibration Results: LDA vs CNN



Calibration recorded	LDA (model C)	CNN two	CNN − LDA
1 trial (≈ 5 min)	91.9	94.3	+2.39 [+1.87, +2.90]
2 trials (≈ 10 min)	92.7	95.2	+2.55
3 trials (≈ 15 min)	93.1	95.9	+2.80

SETUP

CNN checkpoint: 12 gestures scored and mapped to 6 groups, held windows; 19 people × 5 rounds × 3 seeds.

RESULT

Calibrated CNN "two" ahead by +2.4 pp for all 19 people; CNN "small" level with the LDA (+0.3).

BAR NOT MET

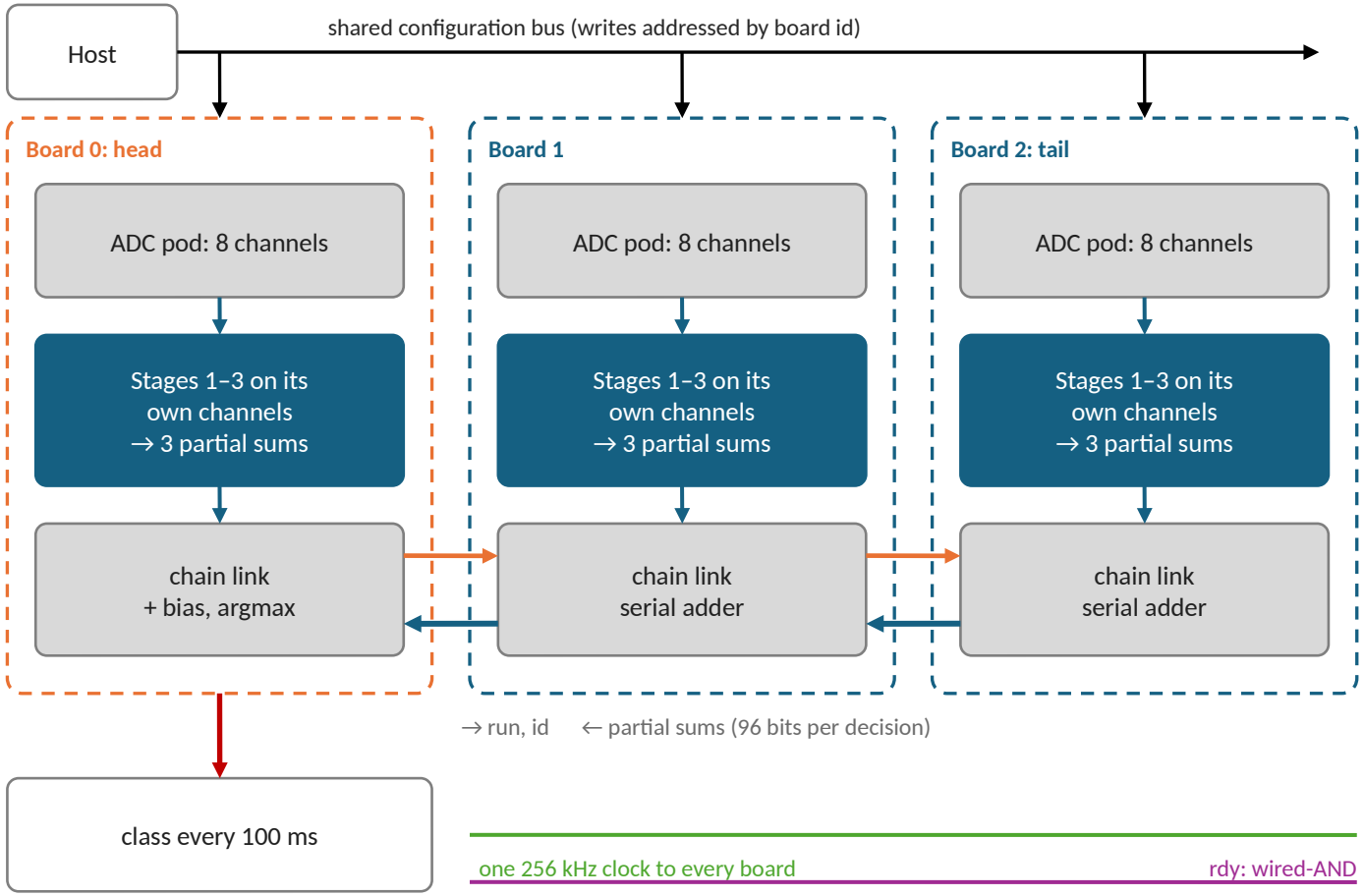
Adoption needed +3 pp; "two" costs ≈ 870× the multiply-adds, a ≥ 9 MHz clock and retraining ≈ 12,000 weights per user.

ARM POSITION

Calibrated in one position: LDA −12.1 pp, CNN −9.6 pp. The CNN is more robust; both need all positions.

We keep the LDA: the calibrated CNN is better for everyone, but by 2.4 pp for ≈ 870× the arithmetic.

From One Chip to a Chain of Boards



BOARD

One 8-channel ADC pod + one chip; chain 1–3 boards for 8, 16 or 24 electrodes.

LOCAL WORK

Each board computes features, normalisation and its share of the LDA on its own channels.

TRAFFIC

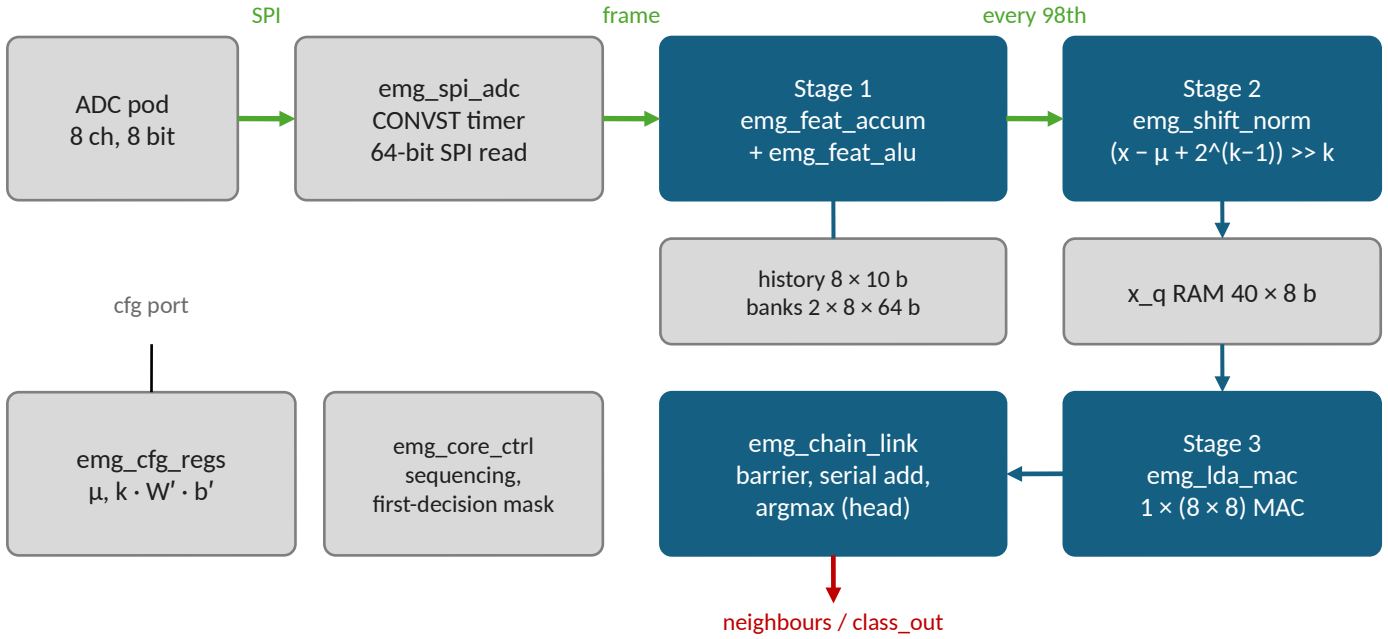
Only 3 partial sums (96 bits) per decision cross between boards.

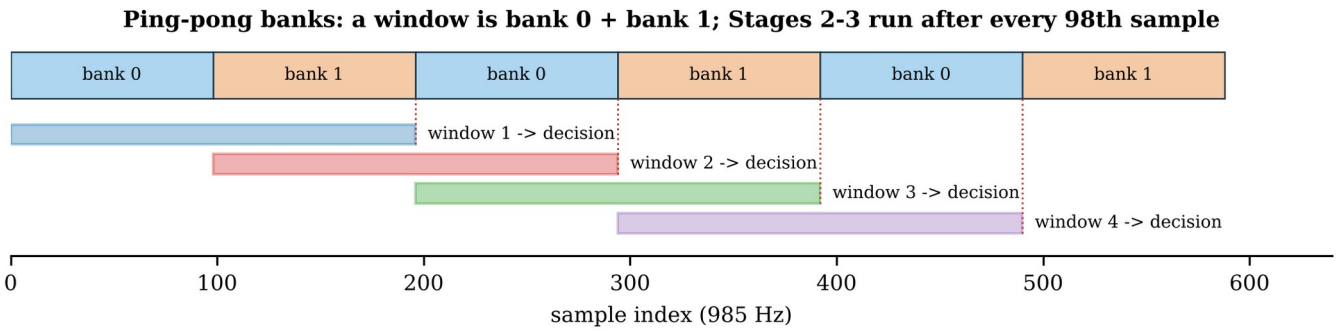
WHY NOT ONE BIG CHIP

Our first chip served 24 channels from one 1 MHz core: sized for 24 even when 8 are used (13.5 vs 3.7 kbit), with two clocks.

Identical boards, one slow clock, and compute that grows with the number of electrodes.

Datapath





EVERY SAMPLE (984.6 Hz)

SPI reads 8 bytes; Stage 1 updates five running sums per channel, one channel per cycle (8 cycles).

EVERY 98TH SAMPLE

Stage 2 normalises 40 features (42 cycles); Stage 3 runs 3 rows × 40 MACs on one multiplier (125 cycles).

WINDOWS

Two ping-pong banks of 98 samples: window = bank 0 + bank 1, one write pointer, no sliding subtraction.

MEMORY

3.7 kbit, all writable, 1-cycle read: maps to SRAM or flip-flops unchanged.

One multiplier, a few adders and 3.7 kbit of memory per board.

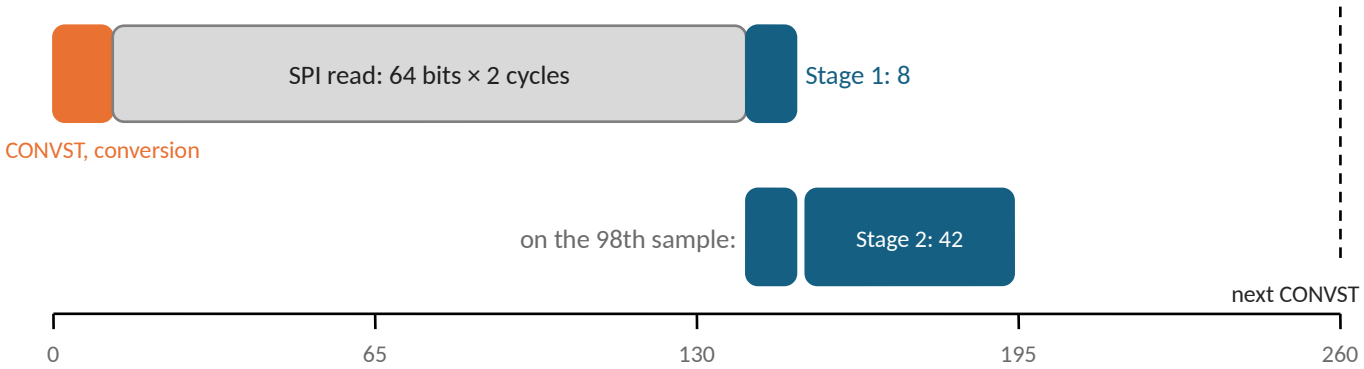
Why Each Choice

Question	Choice	Reason
Window, decision rate	200 ms window, decision every 100 ms	≈ 100 ms to the window centre: inside the 100–125 ms optimum [5]
How to window	2 ping-pong banks of 98 samples	one write pointer; more overlap needs more banks and more decisions
ADC resolution	8 bits	chip model with an 8-bit ADC: no detectable loss (p = 0.49)
Features	add, compare, shift; Mitchell square	no multiplier in Stage 1; non-inferior to float features
Normaliser	(x − μ + 2^(k−1)) >> k, clip to INT8	no divider; no loss against z-scoring
Classifier	K−1 difference rows, one 8 × 8 MAC	120 MACs per 99.5 ms: one multiplier is idle 99 % of the time
Scaling	one INT8 weight scale, INT32 accumulator	argmax unchanged, no requantisation; |score| ≤ 4047 on real data
Argmax	streaming compare, strict >	no comparator tree; ties go to the lower class, deterministic
Constants	writable RAM, not ROM	every electrode layout and every calibrated user needs its own model
First decision	masked (primed)	after reset only one bank is full
Memory size	sized for 8 channels: 3,656 bits	banks 1,024 · μ, k 1,080 · W′ 960 · x_q 320 · b′ 192 · history 80

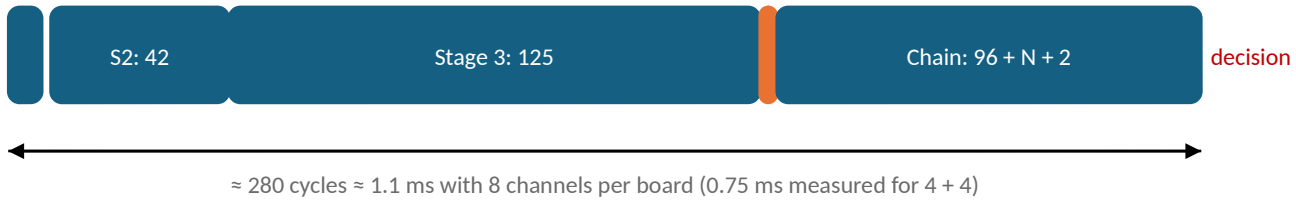
Every block is the cheapest one that costs no measurable accuracy.

One Clock: 256 kHz

One sample period: 260 cycles = 1.016 ms



After the 98th sample (overlaps the next sample's SPI read)



Clock	SPI read per sample	Fits?
128 kHz	1.0 ms of 1.016 ms	no: no room for conversion
256 kHz	**0.5 ms of 1.016 ms**	**yes, half the period free**
1 MHz + 8 MHz (single chip)	16 µs	yes, 4× the core clock and 2 domains

LOWER BOUND

The SPI read: 64 bits at SCLK = clock / 2 takes 128 cycles per sample.

128 kHz

Reading alone would fill the whole 1.016 ms period.

256 kHz

Reading takes half the period; 256 kHz / 260 = 984.6 Hz, within 0.04 % of the data's 985 Hz.

SLACK

A stride's work is ≈ 280 of its 25,480 cycles.

ONE CLOCK

No clock-domain crossing, no drift between boards; 4× lower than the single chip's core clock.

256 kHz is the slowest clock with room to read the ADC; everything else fits easily.

Splitting the LDA: Send K−1 Differences

Score of class c is a sum over features

$$s_c = \Sigma_f W_{cf} q_f + b_c = \Sigma_{boards} \Sigma_{f\ on\ board} W_{cf} q_f + b_c$$

Only the differences to class 0 matter

$$argmax_c\ s_c = argmax\ \{ 0,\ s_1 - s_0,\ s_2 - s_0,\ s_3 - s_0 \}$$

Board j sends three partial sums (the head also adds the bias differences b′)

$$p_c(j) = \Sigma_{f\ on\ board\ j} W'_{cf} q_f, \quad W'_c = W_c - W_0$$

The head decides

$$D_c = \Sigma_j p_c(j), \quad class = argmax\ \{ 0, D_1, D_2, D_3 \}$$

What crosses per decision (one 8-channel board)	Bits
Raw samples (8 ch × 8 bit × 98)	6,272
INT8 features (40 × 8)	320
K = 4 class scores (4 × 32)	128
K − 1 = 3 differences (3 × 32)	**96**

EXACT SPLIT

The LDA score is a sum over features, so it splits across boards without approximation.

DIFFERENCES

argmax only needs each score minus class 0's: class 0 needs no row, and the bias is added once, on the head.

SAVINGS

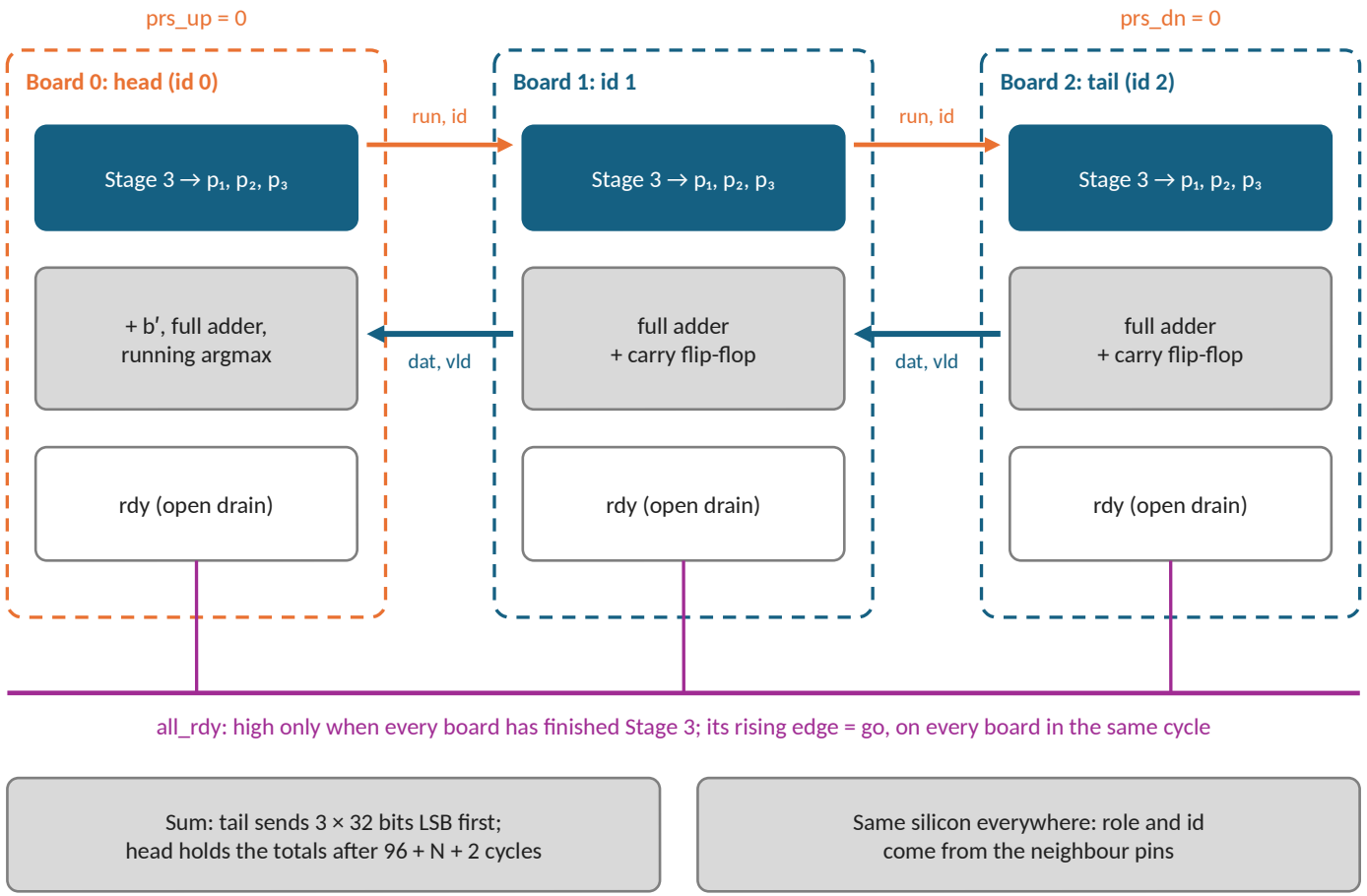
120 instead of 160 MACs and weights; 96 instead of 128 bits per decision.

CHECKED

Same decisions as the single-chip LDA on all 79 strides of real data.

Each board sends 96 bits per decision, whatever its channel count.

Chain Protocol: Who Am I, When to Start, When to Add



1. **POSITION**
   Head = nothing upstream, tail = nothing downstream; the id ripples down (head 0, +1 per board).

2. **CONFIGURATION**
   Shared bus; a write goes to the board whose id equals cfg_board.

3. **START**
   The last write sets the head's core_en; run passes straight down, so every CONVST timer starts in the same cycle.

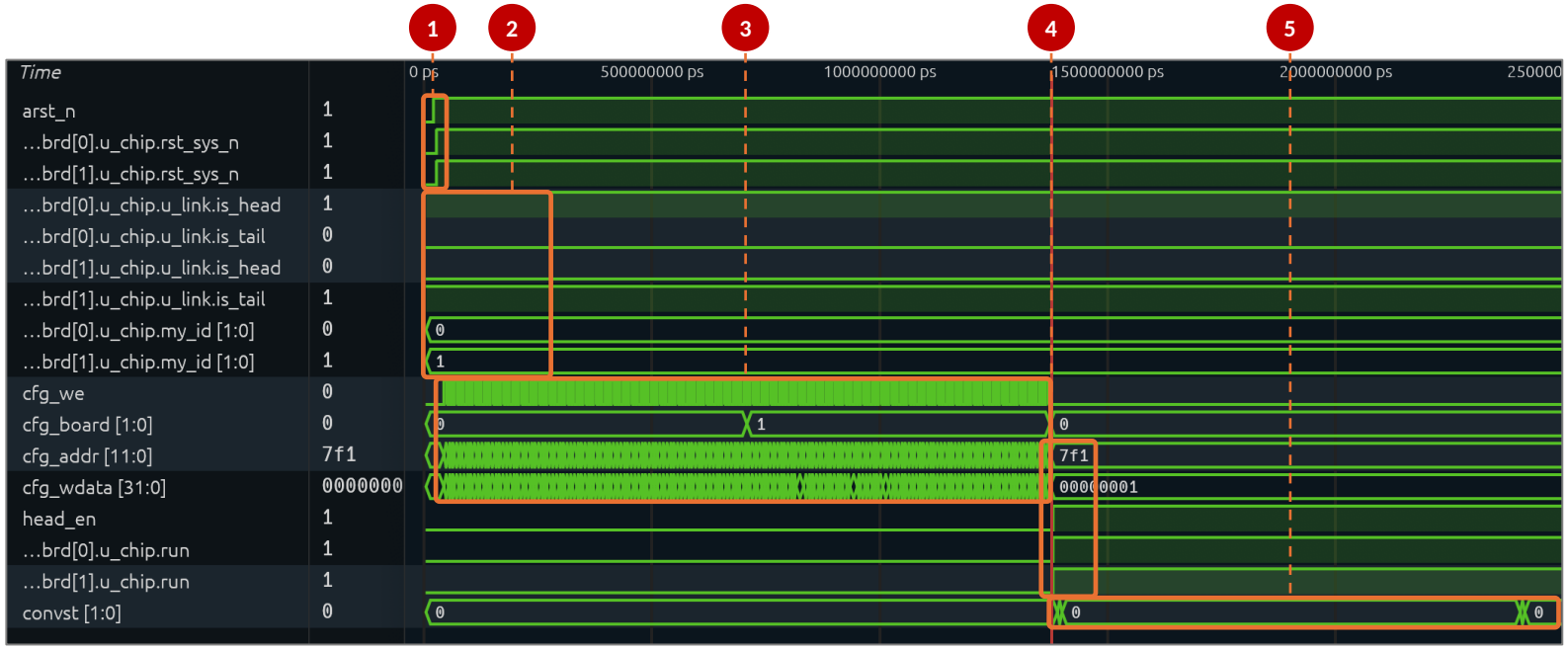
4. **BARRIER**
   The wired-AND rdy waits for the slowest board, with no master polling.

5. **ADD AS THE BITS PASS**
   Each board adds its own bit (one full adder + carry) and forwards it a cycle later; the head compares each total.

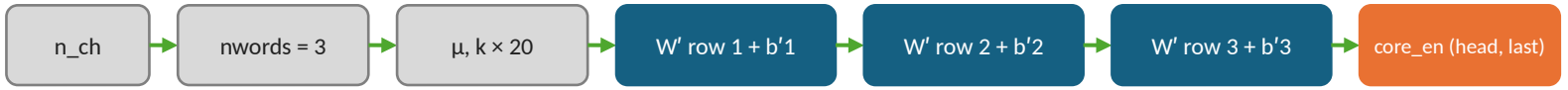
No master and no straps: roles, start and summation all come from the wiring.

OPERATION

Start-up: Reset → Identity → Configuration → Run



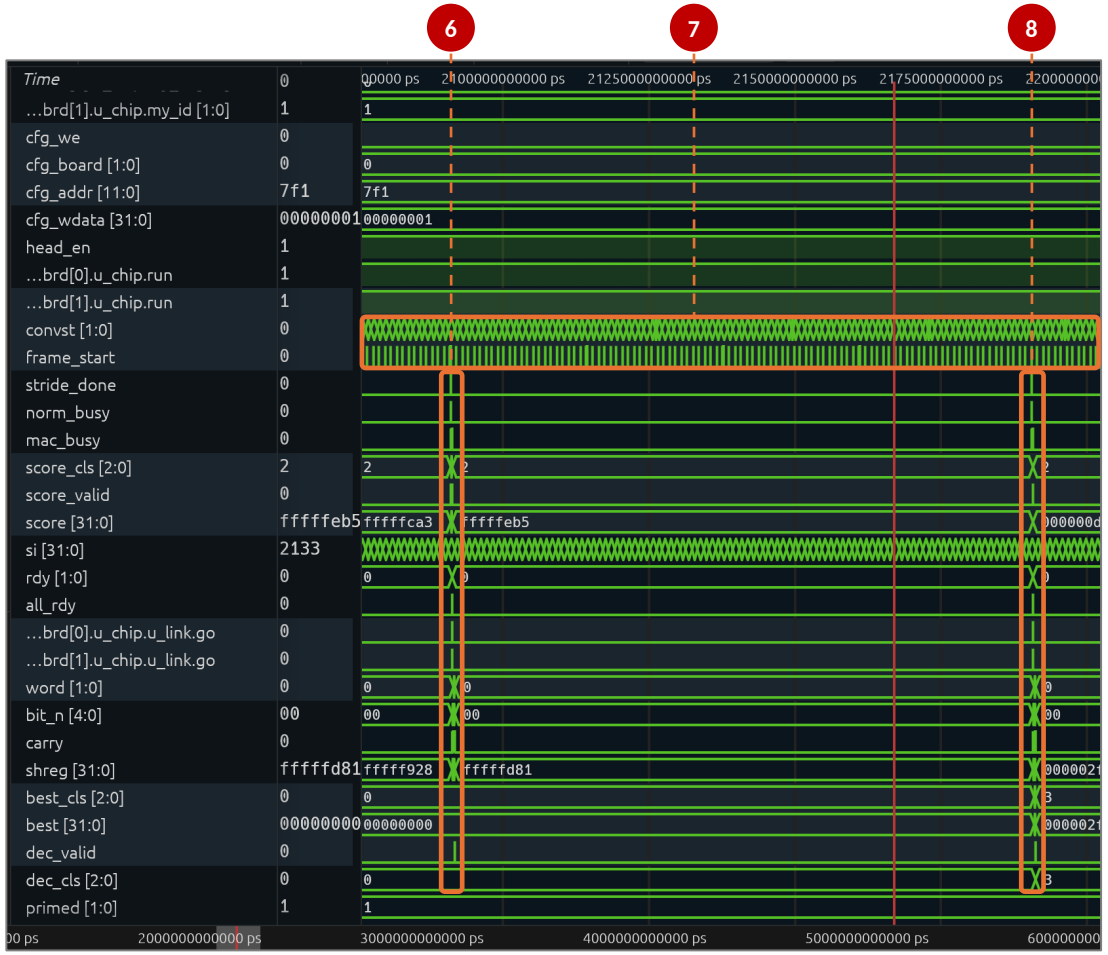
Each board receives, in this order:



1. **Reset**
   arst_n rises at 20 µs; rst_sys_n follows on both boards.
2. **Self-identification**
   From the neighbour pins only: board 0 is_head, board 1 is_tail; my_id = 0, 1.
3. **Configuration**
   171 writes in ≈ 1.34 ms; cfg_board selects the board: 85 to board 0, then 85 to board 1.
4. **Start**
   The last write is the head's core_en; run reaches both boards in the same cycle (1.38 ms).
5. **Lockstep sampling**
   Both CONVST timers start together: one frame every 260 clocks = 1.016 ms.

Both boards find their place, load their model and start sampling together within 1.4 ms.

Steady State: One Burst of Work per Stride

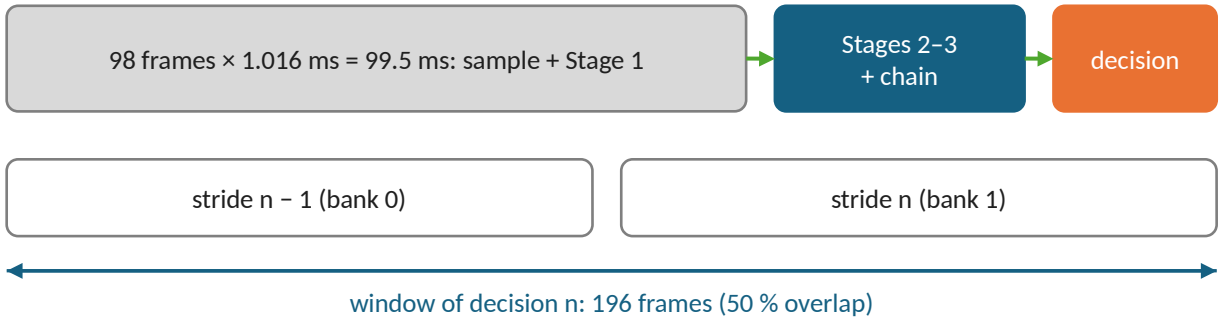


6 Stride ends
After every 98th frame: stride_done → Stage 2 → Stage 3 → rdy / go → chain → dec_valid (decision 19, class 0).

7 Sampling
For the next 98 frames (99.5 ms) only CONVST, the SPI read and Stage 1 run; dec_cls holds the last class.

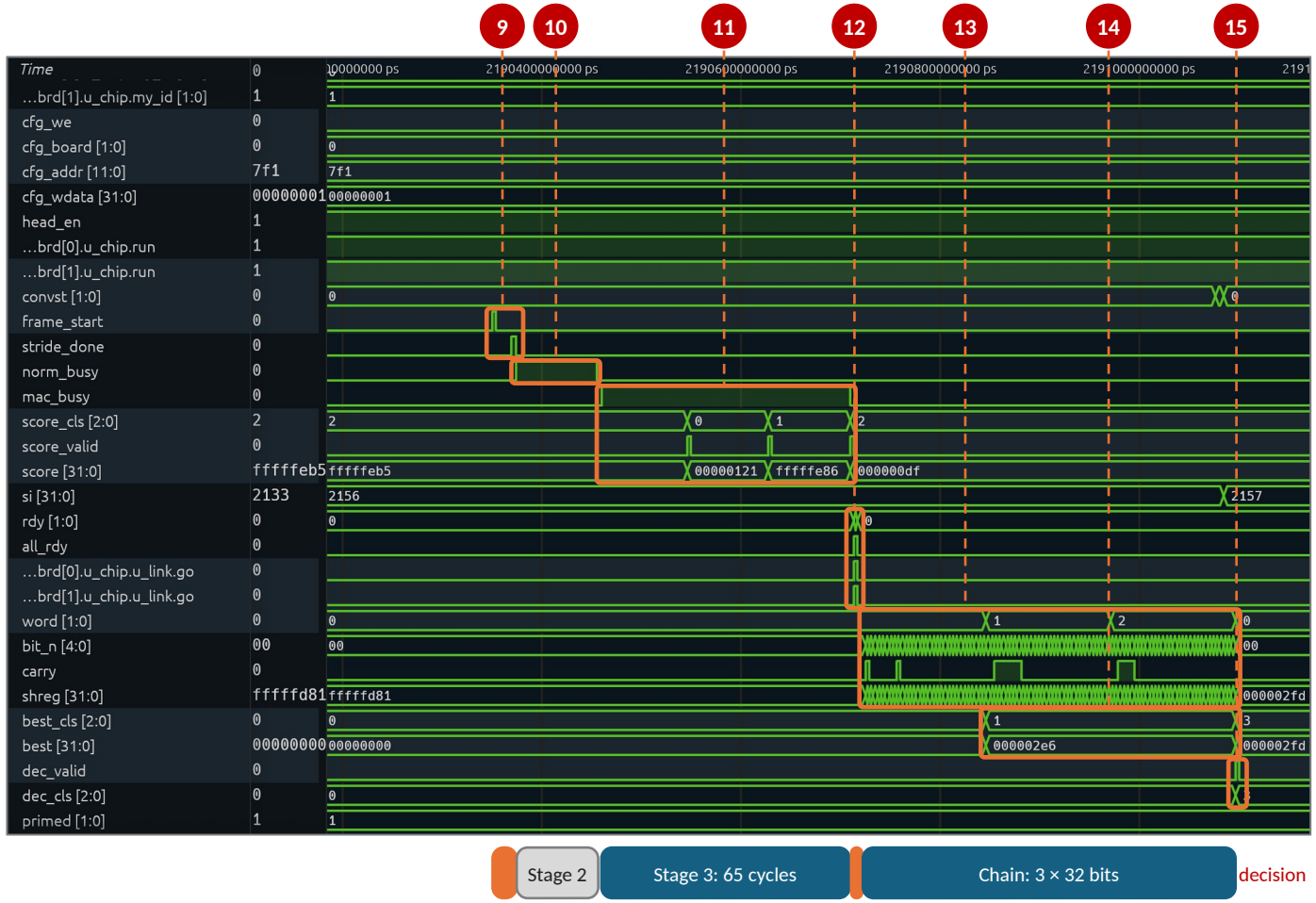
8 Next stride
Decision 20 at 2.191 s: dec_cls 0 → 3. primed = 1: every decision after the first is valid.

One stride (not to scale) and the window of each decision



The boards sample 99 % of the time; one ≈ 0.75 ms burst per stride does all the classification.

Inside One Stride: From the Last Frame to the Decision



9 Last frame
frame_start: the 98th frame is in; Stage 1 updates the features and raises stride_done.

10 Stage 2
norm_busy: 20 features of this board, 22 cycles (86 µs).

11 Stage 3
mac_busy: 3 rows × 21 cycles; partials 289, −378, 223 on score_valid (head, with b′).

12 Barrier
rdy on both boards → all_rdy → go in the same cycle.

13 Chain
The tail streams 3 × 32 bits LSB first; the head adds bit by bit into shreg.

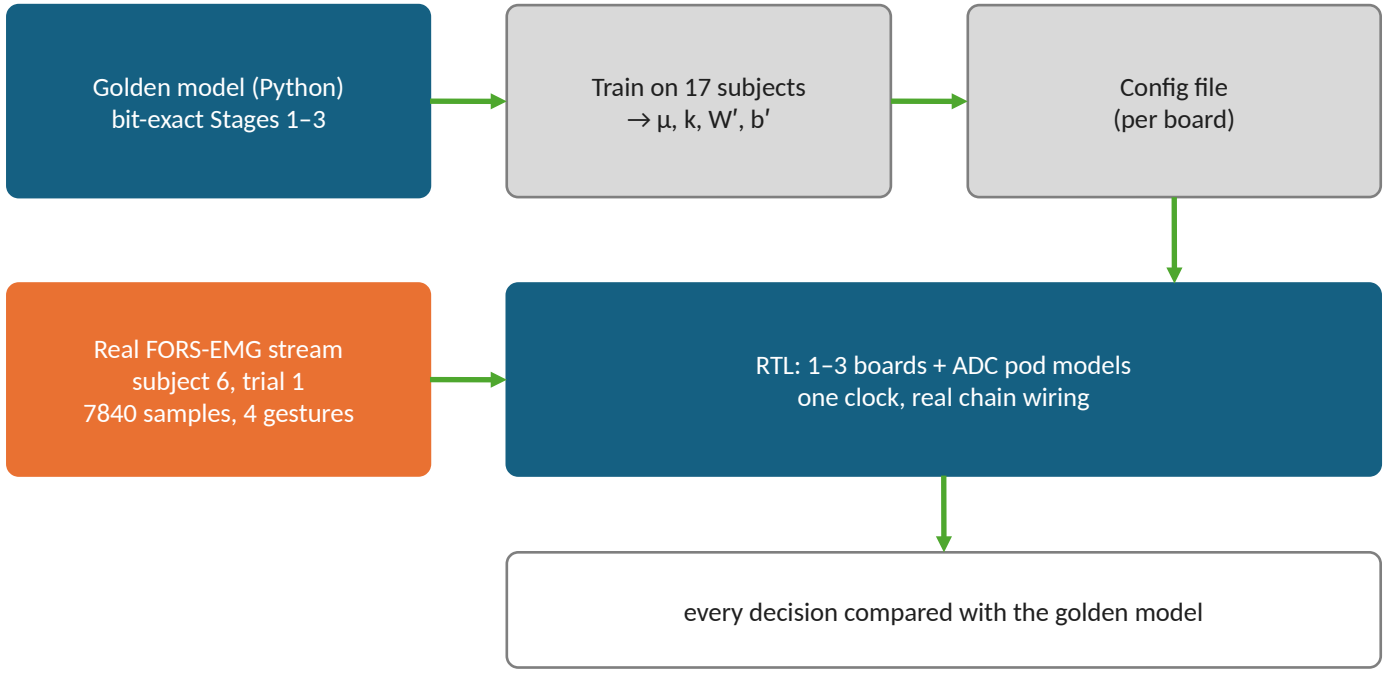
14 Running argmax
D1 = 742 → best_cls 1; D2 = −255; D3 = 765 > 742 → best_cls 3.

15 Decision
dec_valid, dec_cls = 3, ≈ 0.75 ms after the last frame; equals the golden model.

Last frame to decision in ≈ 0.75 ms: D = (742, −255, 765), head + tail, so class 3.

VERIFICATION

Is the Hardware Exactly the Model?



Check	Result
emg_chain_link alone: 1–4 boards × 3000 random strides	pass
8 planted bugs (mutation test)	8 / 8 caught
Whole chip: splits 8, 4+4, 2+6, 6+2, 5+3, 3+3+2	79 / 79 bit-exact
verify_all.py	**18 / 18 pass**

GOLDEN MODEL

Python model of every integer in Stages 1–3; training uses its features, so training and chip see the same numbers.

CHAIN LINK ALONE

1–4 boards × 3000 random strides; expected sums and argmax computed inside the testbench.

MUTATION TEST

8 planted bugs (carry, ≥ instead of >, rdy, ids, …): the test fails on every one.

WHOLE CHIP, REAL DATA

Six board splits: 79 / 79 decisions bit-exact; 72 / 76 single-gesture windows correct.

The hardware is the model, bit for bit.

SUMMARY

Results and Future Work

Step	Result
Gesture set	Hand Close, Hand Open, Wrist Flexion, Wrist Extension: +14.1 pp; merging all 12 instead: −16.9 pp
Model	MLP, logistic regression, SVM, LDA statistically tied; LDA chosen (closed form, 4.3× cheaper)
CNN	Best CNN +2.7 pp; after calibration +2.4 pp; below the +3 pp bar, for ≈ 870× the MACs or more
Arithmetic	INT8: 0.18 pp mean loss; integer features and shift normaliser: no loss
Accuracy, new user	**77.6 % (bit-exact chip model, LOSO); target ≥ 60 %**
With calibration	**86.4 % from one recording per gesture (LDA)**
Hardware	8 channels per board, 1–3 boards, one 256 kHz clock, one multiplier, 96 bits per decision between boards
Verification	79 / 79 decisions bit-exact on real data, 8 / 8 planted bugs caught, 18 / 18 checks

Future work

ON-CHIP CALIBRATION
Update the class means on the device from one recording per gesture.

FOREARM ROTATION
Robustness to the forearm orientation (rest, supination, pronation).

REST / REJECT CLASS
Recognise movements outside the 4 gestures instead of forcing one of them.

SYNTHESIS AND SILICON
Area and power in a target process; tests with real ADC pods.

4 gestures, 77.6 % on a new user (86.4 % calibrated), one multiplier per board, verified bit-exactly.

References

[1] FORS-EMG: A Novel sEMG Dataset for Hand Gesture Recognition Across Multiple Forearm Orientations, arXiv:2409.07484, 2024.

[2] B. Hudgins, P. Parker, R. N. Scott, “A new strategy for multifunction myoelectric control,” IEEE Trans. Biomed. Eng., vol. 40, no. 1, pp. 82–94, 1993.

[3] K. Englehart, B. Hudgins, “A robust, real-time control scheme for multifunction myoelectric control,” IEEE Trans. Biomed. Eng., vol. 50, no. 7, pp. 848–854, 2003.

[4] A. Phinyomark, P. Phukpattaranont, C. Limsakul, “Feature reduction and selection for EMG signal classification,” Expert Syst. Appl., vol. 39, no. 8, pp. 7420–7431, 2012.

[5] T. R. Farrell, R. F. Weir, “The optimal controller delay for myoelectric prostheses,” IEEE Trans. Neural Syst. Rehabil. Eng., vol. 15, no. 1, pp. 111–118, 2007.

[6] J. N. Mitchell, “Computer multiplication and division using binary logarithms,” IRE Trans. Electron. Comput., vol. EC-11, no. 4, pp. 512–517, 1962.

[7] O. Ledoit, M. Wolf, “A well-conditioned estimator for large-dimensional covariance matrices,” J. Multivariate Anal., vol. 88, no. 2, pp. 365–411, 2004.